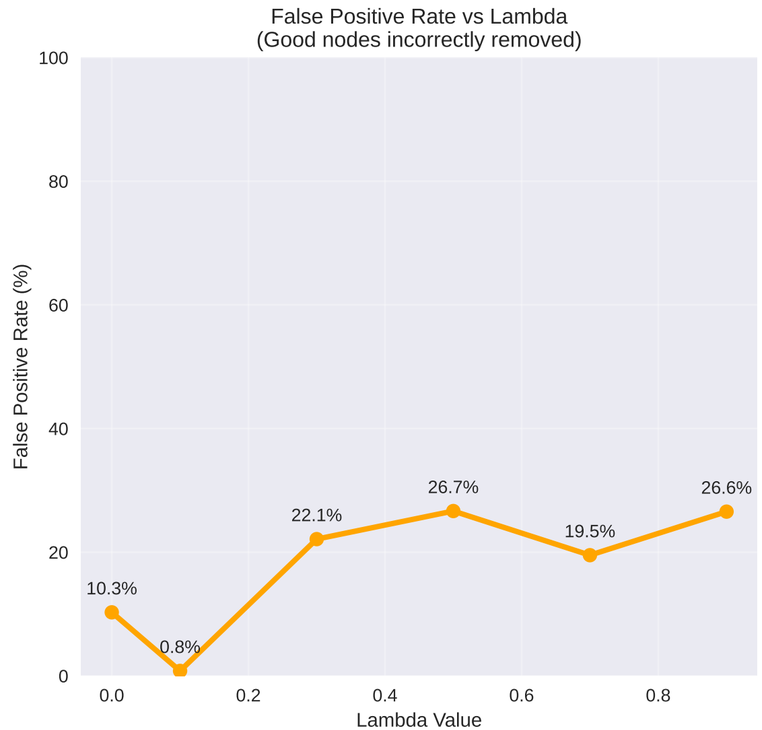
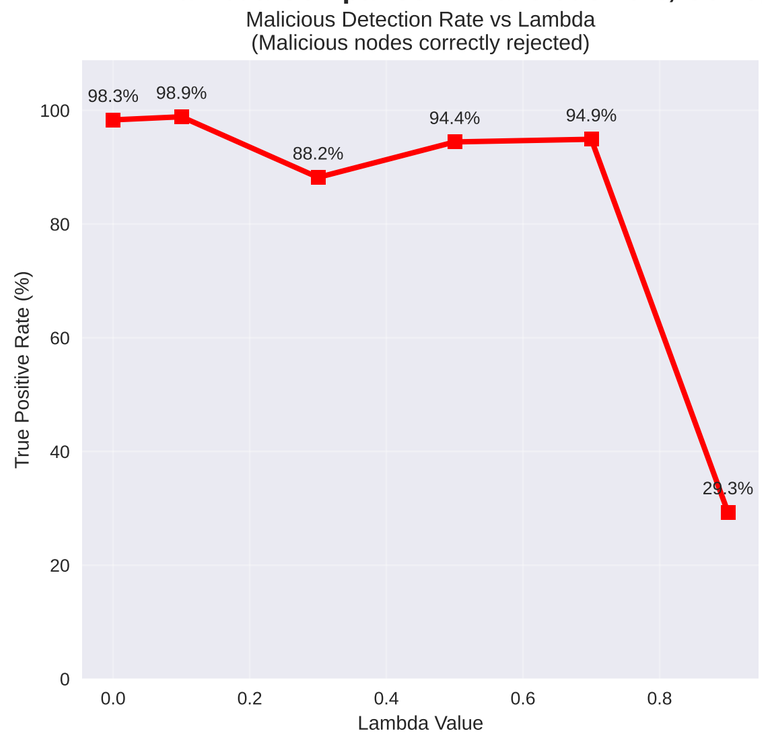
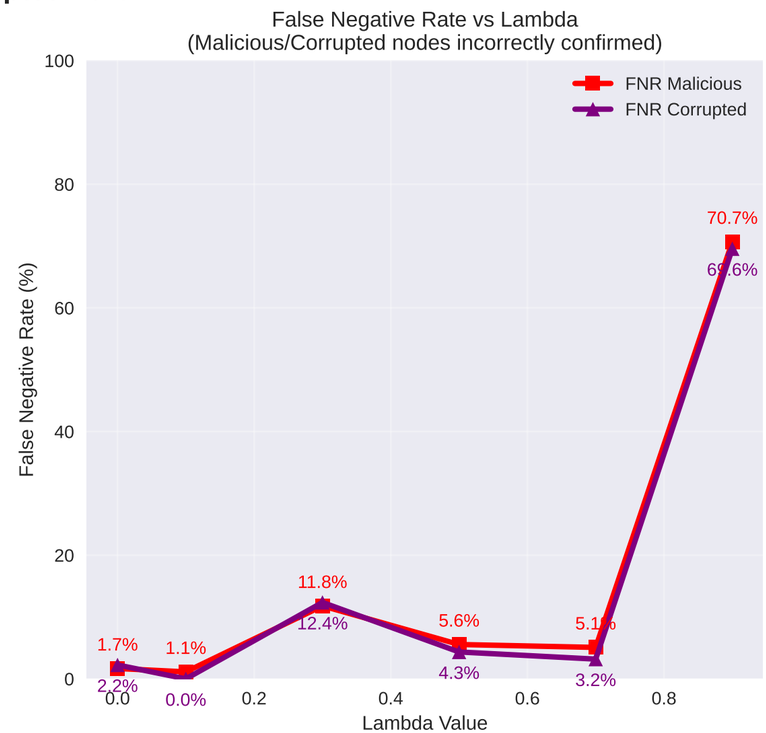
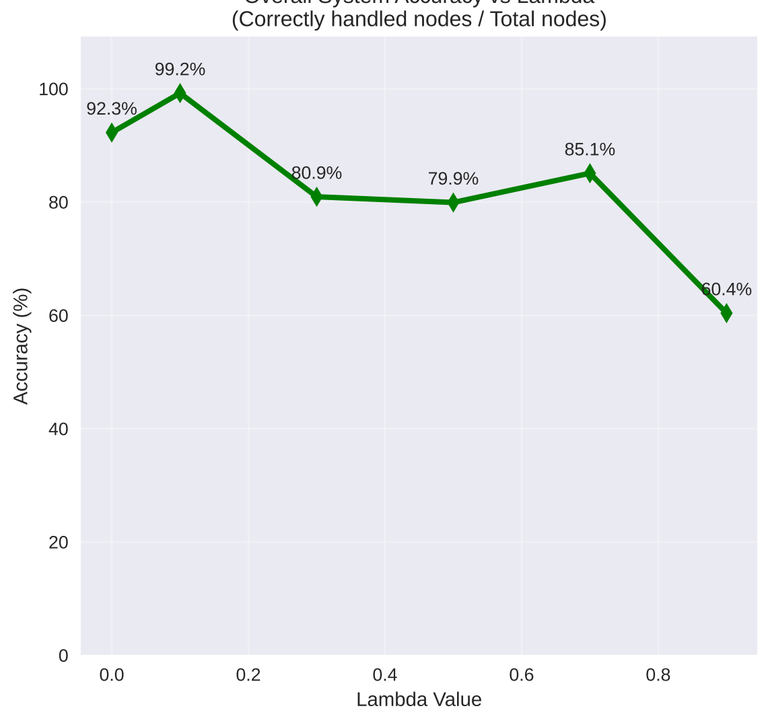
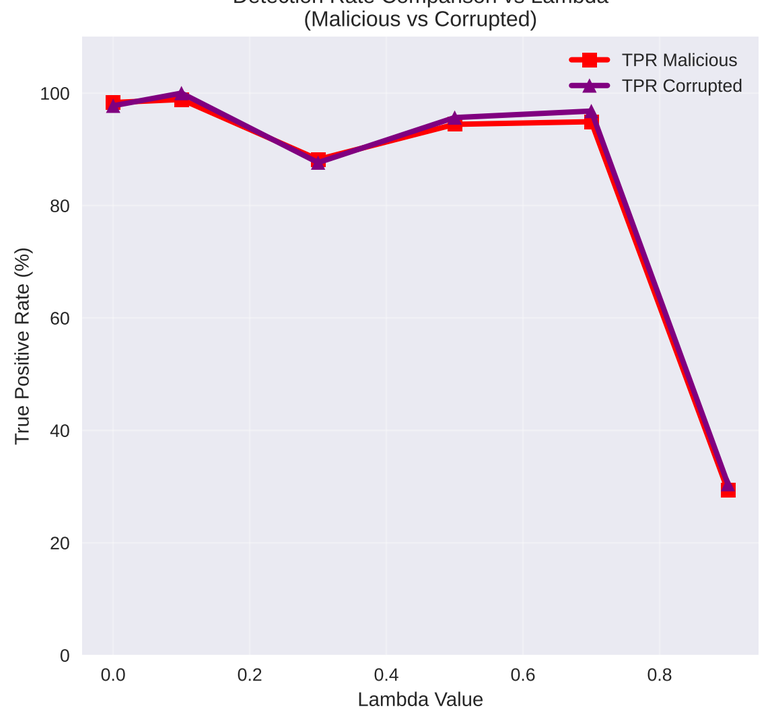
Lambda Comparison - Malicious 20%, Corrupted 10%

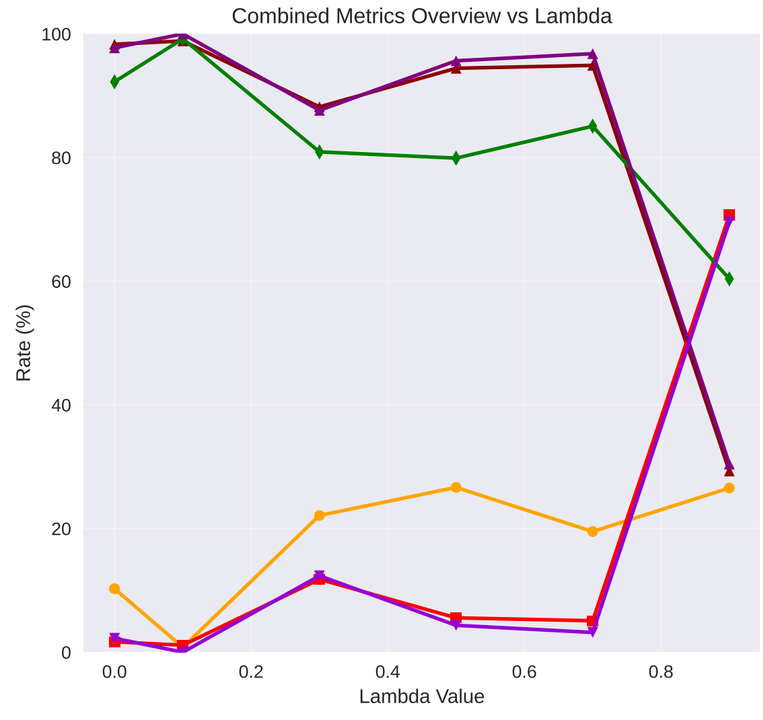












Lambda Effectiveness Ranking

```
LAMBDA EFFECTIVENESS RANKING
========================================

1. Lambda = 0.1
   FPR: 0.8%, TPR: 98.9%, FNR: 1.1%, Acc: 99.2%
   Score: 196.2

2. Lambda = 0.0
   FPR: 10.3%, TPR: 98.3%, FNR: 1.7%, Acc: 92.3%
   Score: 178.6

3. Lambda = 0.7
   FPR: 19.5%, TPR: 94.9%, FNR: 5.1%, Acc: 85.1%
   Score: 155.4

4. Lambda = 0.5
   FPR: 26.7%, TPR: 94.4%, FNR: 5.6%, Acc: 79.9%
   Score: 142.1

5. Lambda = 0.3
   FPR: 22.1%, TPR: 88.2%, FNR: 11.8%, Acc: 80.9%
   Score: 135.2

6. Lambda = 0.9
   FPR: 26.6%, TPR: 29.3%, FNR: 70.7%, Acc: 60.4%
   Score: -7.6


LAMBDA PARAMETER:
The lambda parameter controls the weight of historical trust
information in the TrustInformedTipSelector algorithm.
Higher lambda values give more weight to past behavior.

LAMBDA VALUES TESTED:
Values: [0.0, 0.1, 0.3, 0.5, 0.7, 0.9]
Configuration: Malicious 20%, Corrupted 10%
```

Summary Statistics

```
LAMBDA COMPARISON ANALYSIS - SUMMARY
====================================

Generated: 2025-08-31 14:55:19
Results Path: FL/results_lambda

CONFIGURATION ANALYZED:
• Malicious: 20%, Corrupted: 10%
• Lambda values tested: [0.0, 0.1, 0.3, 0.5, 0.7, 0.9]

DETAILED RESULTS:
\n• Lambda 0.0:\n  FPR=10.3%, TPR=98.3%, FNR=1.7%, Acc=92.3%\n\n• Lambda 0.1:\n  FPR=0.8%, TPR=98.9%, FNR=1.1%, Acc=99.2%\n\n• Lambda 0.3:\n  FPR=22.1%, TPR=88.2%, FNR=11.8%, Acc=80.9%\n\n• Lambda 0.5:\n  FPR=26.7%, TPR=94.4%, FNR=5.6%, Acc=79.9%\n\n• Lambda 0.7:\n  FPR=19.5%, TPR=94.9%, FNR=5.1%, Acc=85.1%\n\n• Lambda 0.9:\n  FPR=26.6%, TPR=29.3%, FNR=70.7%, Acc=60.4%\n

METRICS EXPLANATION:
• FPR (False Positive Rate): Good nodes incorrectly removed / Total good nodes sent
• TPR (True Positive Rate): Malicious nodes correctly detected / Total malicious nodes sent
• FNR (False Negative Rate): Malicious nodes incorrectly confirmed / Total malicious nodes sent
• Accuracy: Overall system accuracy (correctly handled nodes / total nodes)

CALCULATION METHOD:
• Each round selects K-1 = 9 nodes randomly from K = 10 peers
• Uses actual worker selection data from 3000_workers_selected.csv
• Sums Good_Nodes_Removed_This_Round and Malicious_Nodes_Confirmed_This_Round across ALL rounds
• Counts actual occurrences of each peer type in selected workers

OPTIMAL VALUES:
• FPR: 0% (no good nodes removed)
• TPR: 100% (all malicious nodes detected)
• FNR: 0% (no malicious nodes confirmed)
• Accuracy: 100% (perfect classification)

LAMBDA PARAMETER:
The lambda parameter controls the weight of historical trust information
in the TrustInformedTipSelector algorithm. It determines how much influence
past behavior has on current trust decisions:
• Lambda = 0.0: No historical influence (only current round)
• Lambda = 1.0: Maximum historical influence
• Higher lambda values may improve detection but can be slower to adapt
```